Hand Gesture Recognition for ASL

Trained Completely From Scratch — No Pretrained Models

A Custom CNN achieving **99.24% Validation Accuracy** on 32 ASL Classes at **14–18 FPS on CPU**

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Abstract

A complete hand gesture recognition system built **entirely from scratch** using a custom CNN — no pretrained weights, transfer learning, or external APIs. Recognises **32 ASL gestures** achieving **99.24% best validation accuracy**. Dataset of 12,800 images self-collected via custom webcam pipeline. Runs at **14–18 FPS on CPU**.

Keywords: CNN · ASL · Custom Dataset · OpenCV · TensorFlow

Dataset

Name	Date modified	Type
A	27-04-2026 04:03	File folder
B	27-04-2026 04:03	File folder
bye	27-04-2026 04:14	File folder
C	27-04-2026 04:03	File folder
D	27-04-2026 04:04	File folder
E	27-04-2026 04:04	File folder
F	27-04-2026 04:05	File folder
G	27-04-2026 04:05	File folder
H	27-04-2026 04:06	File folder
hello	27-04-2026 04:14	File folder
I	27-04-2026 04:06	File folder
J	27-04-2026 04:07	File folder
K	27-04-2026 04:07	File folder
L	27-04-2026 04:08	File folder
M	27-04-2026 04:08	File folder
N	27-04-2026 04:09	File folder
no	27-04-2026 04:14	File folder
O	27-04-2026 04:09	File folder

- **32 classes:** A–Z + hello, bye, yes, no, thankyou, perfect
- **400 images/class:** 200 left + 200 right hand
- **12,800 total:** 80% train / 20% val
- ROI-box — only hand region saved, no face
- All images captured by **Riyan Ahlawat**

Methodology

1. **Collect:** ROI webcam script; left + right hand
2. **Preprocess:** Resize 64×64; normalise [0,1]; tf.data
3. **Augment:** Flip, brightness, contrast, hue
4. **Model:** 4 conv blocks 32→64→128→256; GAP; Dense
5. **Train:** Adam + ReduceLROnPlateau + EarlyStopping
6. **Infer:** OpenCV; every 2 frames; top-3 overlay

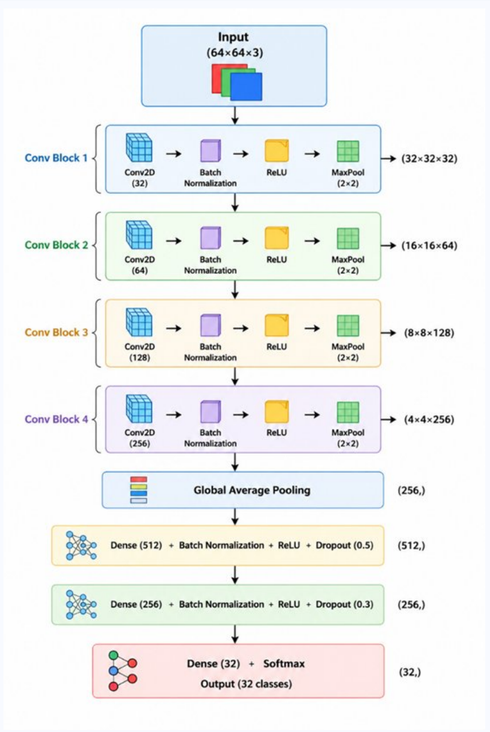
Challenges

- **Python 3.14 —** TF needs ≤3.12; installed 3.11
- **Domain gap —** black-bg dataset failed; rebuilt from webcam
- **GPU on Windows —** TF ≥2.11 no native GPU; used CPU
- **Laptop shutdown —** sleep disabled; charger connected
- **Save format —** .keras incompatible TF2.13; used .h5

Team

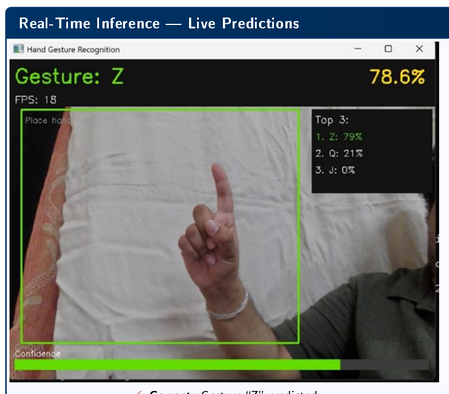
Member	Role
Riyan Ahlawat	Data collection, pipeline, training
Shivam Rajawat	CNN architecture, tuning
Samarth Jakhmola	Inference, OpenCV integration
Pratham Bhat	Evaluation, reporting

CNN Architecture

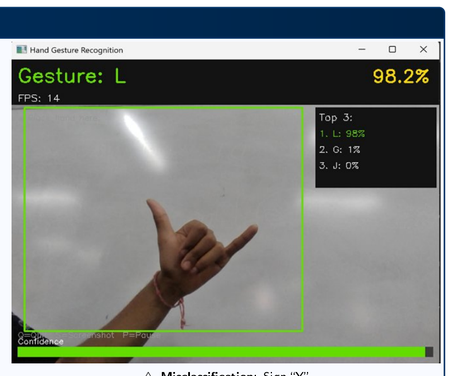


He Normal init · L2 reg (10^{-4}) · BatchNorm every layer · Global Avg Pool · Dropout 0.5/0.3 · **664,544 params**

Real-Time Inference — Live Predictions



✓ **Correct:** Gesture "Z" predicted at 78.6% confidence (18 FPS)



△ **Misclassification:** Sign "Y" predicted as "L" (98.2%)

Results

Metric	Value
Best Val Accuracy	99.24%
Final Val Accuracy	97.11%
Final Val Loss	0.1620
Best Epoch	34 / 35
Train Accuracy	100%
Speed (CPU)	14–18 FPS
Model Size	∼2.5 MB
Parameters	664,544

35 epochs, batch 32. Best val accuracy **99.24%** at epoch 34. EarlyStopping prevented overfitting. Final train accuracy: **100%**.

Key Contributions

- CNN built **from scratch** — zero pretrained knowledge
- 12,800 self-collected **bilateral webcam images**
- Purpose-built **ROI data collection tool**
- Real-time **14–18 FPS inference** on CPU
- Domain gap solved by matching train/inference environment

Conclusion

Custom CNN trained from scratch achieves **99.24% best val accuracy** on 32 ASL classes at **14–18 FPS on CPU**. No pretrained weights or APIs used. Domain gap — original black-background dataset failing on webcam — solved by rebuilding dataset in the real environment.

Future: dynamic gestures, mobile deployment.

Hyperparameters

Parameter	Value
Optimizer	Adam ($\eta = 0.001$)
Loss	Sparse Cat. Cross-Entropy
Epochs	35
Batch Size	32
LR Decay	$\times 0.5$ (patience 5)
Early Stop	patience 10
Weight Init	He Normal
L2 Reg	10^{-4}
Dropout	0.5 / 0.3
Input	$64 \times 64 \times 3$

Applications

- **Accessibility:** ASL communication for deaf/HoH community
- **HCI:** Touchless computer interfaces
- **Education:** Interactive ASL learning tools
- **Healthcare:** Gesture-based patient communication
- **Gaming/AR:** Controller-free immersive input

Training Configuration

- tf.data pipeline with prefetching
- Data augmentation: random flip, brightness (± 0.2), contrast (± 0.2), hue (± 0.05)
- He Normal weight initialisation
- BatchNorm after every Conv and Dense layer
- Global Average Pooling replaces Flatten to reduce params
- ReduceLROnPlateau: monitor val_loss, patience 5
- EarlyStopping: monitor val_accuracy, patience 10
- ModelCheckpoint: saves best .h5 weights

Reproducibility & Stack

Component	Version/Detail
Framework	TensorFlow 2.13
Language	Python 3.11
Inference	OpenCV 4.x
Hardware	CPU only (no GPU)
Model file	.h5 format
Dataset	12,800 images / 32 classes

No external APIs. Fully CPU-deployable.

Model Architecture Summary

- **Conv Block 1:** Conv2D(32) → BN → ReLU → MaxPool ⇒ 32×32×32
- **Conv Block 2:** Conv2D(64) → BN → ReLU → MaxPool ⇒ 16×16×64
- **Conv Block 3:** Conv2D(128) → BN → ReLU → MaxPool ⇒ 8×8×128
- **Conv Block 4:** Conv2D(256) → BN → ReLU → MaxPool ⇒ 4×4×256
- **GAP** ⇒ (256,)
- **Dense(512)** + BN + ReLU + Dropout(0.5)
- **Dense(256)** + BN + ReLU + Dropout(0.3)
- **Dense(32)** + Softmax ⇒ Output

Total params: **664,544** | Size: ∼**2.5 MB**

Training History

35 epochs · Batch 32 · Adam with ReduceLROnPlateau. Best val accuracy **99.24%** at epoch 34. EarlyStopping active. Train accuracy: **100%**.

Architecture at a Glance

4 Conv Blocks (32→64→128→256) + GAP + Dense(512) + Dense(256) + Softmax(32). Total: **664,544 params**. Model: ∼**2.5 MB**. Input: $64 \times 64 \times 3$.

Impact & Future Work

Zero pretrained knowledge. Real-time CPU inference at 14–18 FPS. Domain gap solved. Future: **dynamic gestures**, **mobile deployment**, larger ASL vocabulary.